Gaps in the interpretation of pronouns

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SALT 28

Motivations

Puzzle: universal/existential reading of donkey anaphora

(1) a. Every farmer who owns a donkey$_1$ cherishes it$_1$.
$\leftrightsquigarrow$ cherishes **all** of his donkeys.

b. Every customer with a credit card$_1$ pays with it$_1$.
$\leftrightsquigarrow$ pays with **one** credit card.

Questions:

- How does the pronoun access its referent?
- Where does the ambiguity originate?

Results & proposal

A. The readings surface in non-quantified environments, e.g. simple conjunctions.

B. The ambiguity is local to the pronoun, i.e. implicit quantification takes narrow scope.

C. The ambiguity arises when the pronoun must choose between multiple referent

D.

Type-shifting accounts

(Rooth, 1985; Root, 1986; Groenendijk et al., 1991; Champollion et al., 2017)

Account:

- Scope-extension rule for cross-clausal anaphora:
  "Some A$_1$ is B; it$_1$ is C" $\leadsto \exists x : A(x), (B(x) \wedge C(x))$
- Two recipes for creating dynamic quantifiers:

(2) $Q(\text{restrictor})(\text{scope})$

a. Universal: $Q(\text{restrictor})\,(\text{restrictor} \wedge \text{scope})$

b. Existential:
$Q(\text{restrictor})\,(\neg\,(\text{restrictor} \wedge \neg\text{scope}))$

Consequence: the validity of the type-shifting accounts is tied to the validity of the *scope-extension rule*.

Figure: Figure caption

#1: Narrow pseudo-scope reading

(3) a. Every citizen who has a valid ID$_1$ shows it$_1$ whenever she is asked to.

b. Every father who received a chocolate bar$_2$ gave a piece of it$_2$ to each of his kids.

Chance 1 Chance 2 Chance 3

Reading: $\forall \gg$ whenever $\gg \exists$

Weakest possible reading under the type-shifting accounts:

$\forall \gg \exists \gg$ whenever

#2: Universal readings across conjunctions

Bets:

- Winning conditions = Truth-conditions
- No scalar implicatures in their scope

(4) I bet you 10$ that some students will fail the test.

All students fail $\Rightarrow$ Speaker wins

54

zefjopzefop

6

zefjopzefop

Analysis

I. E-type pronoun (Cooper, 1979)[a] + Vagueness

(5) *it* $\leadsto \{f\,(\text{donkey he owns}) \mid f \text{ is a choice function}\}$

II. Vagueness projects as in alternative semantics (Kratzer et al., 2017).

(6) a. $[\![\mathbf{f}\ \mathbf{x}]\!]^{g,w} = \{f(x) \mid f \in [\![\mathbf{f}]\!]^{g,w}, x \in [\![\mathbf{x}]\!]^{g}\}$

b. $[\![\lambda_i\ \mathbf{p}]\!]^{g,w} = \left\{f \in D_{et} \;\middle|\; \forall x,\ f(x) \in [\![\mathbf{p}]\!]^{g[i/x],w}\right\}$

$\leadsto$ different values for i may use different alternatives.

III. Trivalent convention: $\# := \{\text{true}, \text{false}\}$

IV. Križ's maxim of quality for trivalent propositions (using an idea of Champollion, 2016)
S true enough in w with respect to issue I
iff $\exists w'$, S is true in w' and $w \approx_I w'$

Derivation

Cross-conjunction anaphora

(7) a. A donkey came and ...

b. ... it[donkey that came] brayed

$$[\![(\mathbf{??})]\!]^{g,w} = \left\{\texttt{bray}'\,(f\,(\text{donkey that came})) \;\middle|\; f \in CH\right\}$$
$$= \begin{cases} \textbf{true} & \text{if all donkeys that came brayed} \\ \textbf{false} & \text{if no donkeys that came brayed} \\ \# & \text{otherwise} \end{cases}$$

Simple donkey sentences

(8) Every farmer that has a donkey λ_1. t_1 cherishes it[donkey he$_1$ owns]$_\alpha$.

$$[\![\alpha]\!]^{g,w} = \left\{g(1) \text{ cherishes } f\left(\texttt{doneky-owned-by}'(g(1))\right) \;\middle|\; f \in CH\right\}$$
$$[\![(\mathbf{??})]\!]^{g,w} = \begin{cases} \textbf{true} & \text{if all donkey-owning farmers cherishes all donkey they own} \\ \textbf{false} & \text{if some donkey-owning farmer cherishes no donkey he owns} \\ \# & \text{otherwise} \end{cases}$$

Conclusion

Nunc tempus venenatis facilisis. **Curabitur suscipit** consequat eros non porttitor. Sed a massa dolor, id ornare enim. Fusce quis massa dictum tortor **tincidunt mattis**. Donec quam est, lobortis quis pretium at, laoreet scelerisque lacus. Nam quis odio enim, in molestie libero. Vivamus cursus mi at *nulla elementum sollicitudin*.

[a] A dynamic approach is also conceivable.

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